Glass Stats and Crime: Let's Make the Right Decision

F. Jimenez[1,2], A. Koepke[2], R. Corzo[2], E. Steel[2]

[1]University of Colorado Boulder [2]National Institute of Standards and Technology

July 31, 2019

Outline

Is this Glass Fragment from the Crime Scene?

Evidence Collected

- Glass fragments (K) from a known source. Collected at a crime scene.
- A single glass fragment (Q) from an unknown source. Found on a suspect.

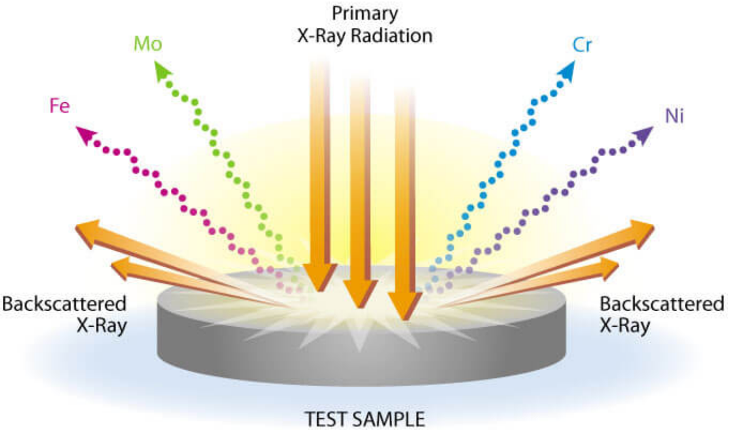
Question We Need to Answer

Does the single questioned fragment (Q) come from the same source as the collection of known fragments (K)?

Motivation

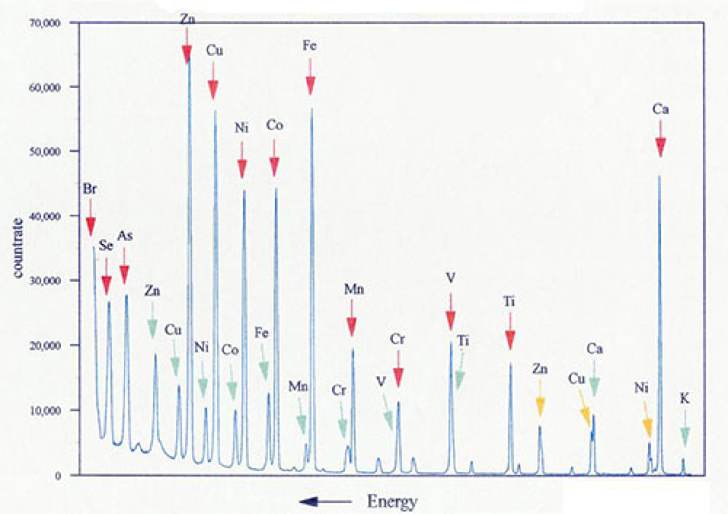
Destructive methods already exist, but we want to save the evidence. Can we use μ-XRF?

Basic Idea Behind μ-XRF



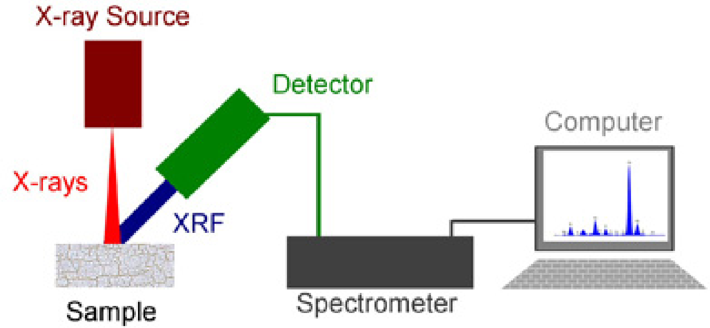
X-Rays are focused on a sample and secondary radiation is released

μ-XRF Spectrum



μ-XRF spectrum

Measurement Setup

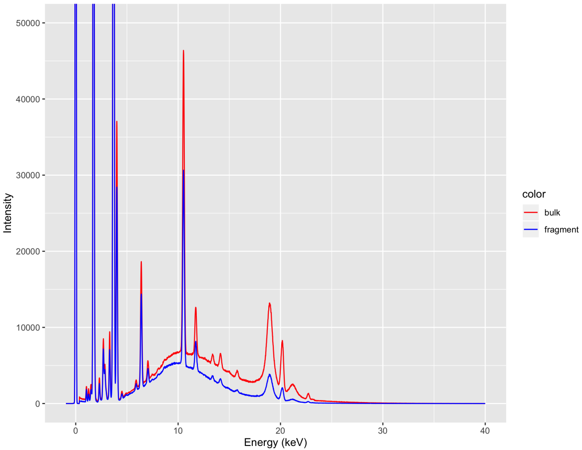


The entire data collection process.

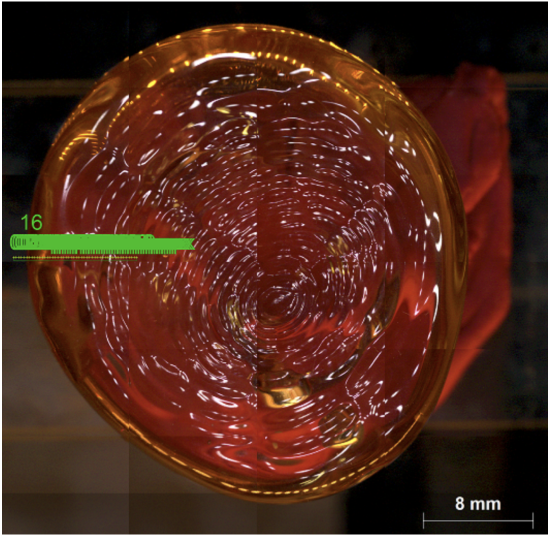
Changing Data

Causes of Differences

- Settings on machine
- Variations in shape and sample thickness

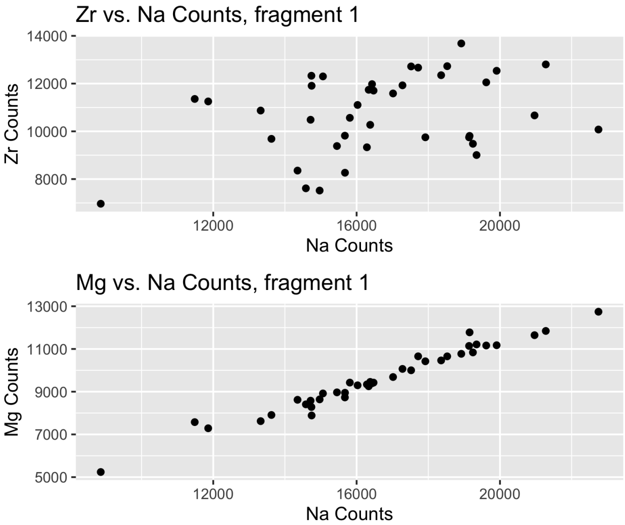


Varying Measurement Location



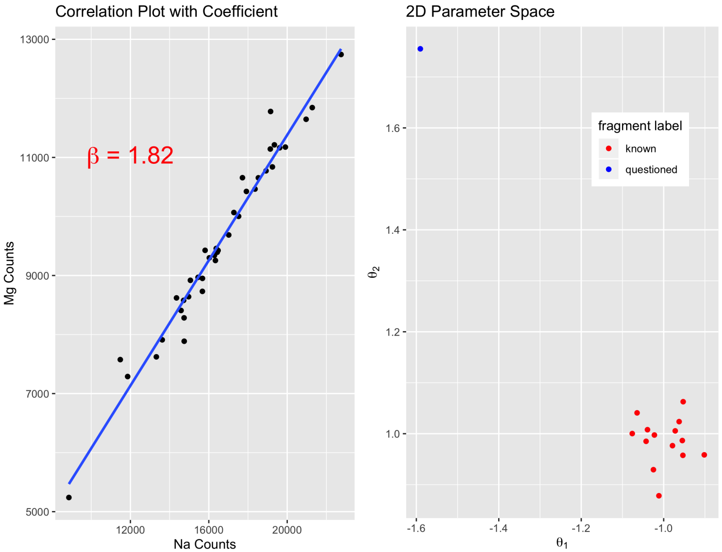
Locations where a measurement were made.

Correlation Plots



Correlation plots for elements with similar atomic numbers will be linear

Parameter Space

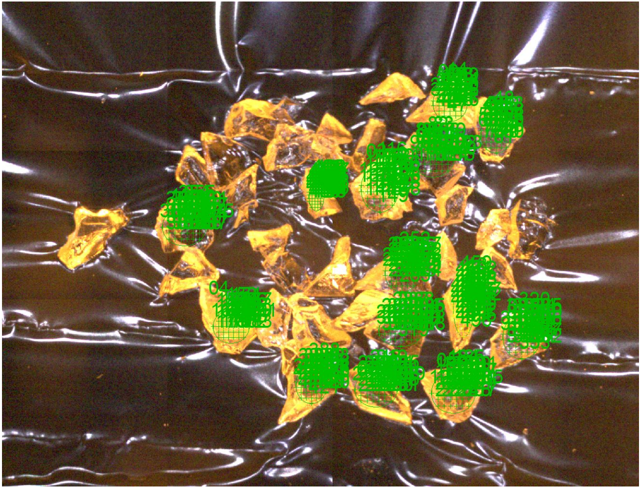


Testing for the Same Source

Permutation Test

- The labels Q and K are arbitrary
- Our test statistic will be the Mahalanobis distance between the parameters of both groups.
- We cannot permute individual measurements, but instead we permute entire fragments.

House Window Example

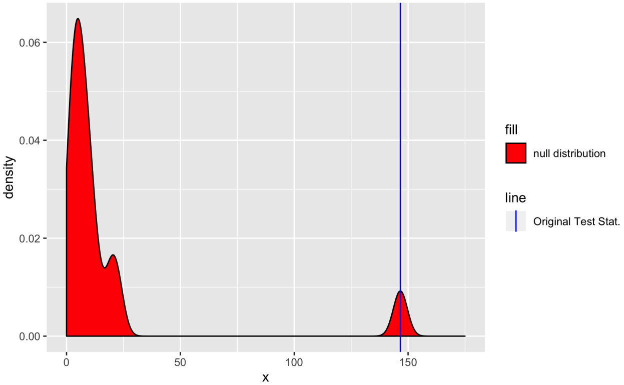


Glass fragments from a house window measured at the locations in green

House Window Example Results

Null distribution and original test statistic when **sources match**

House Window Example Results



Null distribution and original test statistic when **sources don't match**

Final Thoughts

Limitations of the Current Test

- The null hypothesis is same source, but this doesn't match the way evidence is supposed to be presented
- Significance level α is $\frac{1}{N}$
- Analyzing more than 10-20 fragments is not feasible

Value of Collaboration

- Input from the domain experts was critical for coming up with our data processing procedure
- Providing feedback to the domain expert on what they can expect from small samples can help guide their work

Conclusion

Summary

- μ-XRF introduction
- Data processing, guided by scientists
- Permutation test on real data
- Limitations of our method
- Value of collaboration